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First in vivo images of the human brain revealed with the Iseult 11.7T MRI scanner

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Abstract

The understanding of the human brain is one of the main scientific challenges of the 21st century. In this context, in the early 2000s the French Atomic Energy Commission (CEA) launched a program to conceive and build the first human MRI scanner operating at 11.7T. More than a decade of developments followed to deliver the magnet while six more years were necessary to complete the commissioning and finally obtain approval from the regulatory agencies to acquire the first ever human brain images *in vivo* at such magnetic field. We deployed parallel transmission tools to mitigate the radiofrequency field inhomogeneity problem and tame the specific absorption rate. To assure the safety of human imaging at such high field strength, we performed physiological, vestibular, behavioral and genotoxicity measurements on the volunteers. The data shows no evidence of adverse effects. The unprecedented field strength combined with the acquisition techniques deployed yield T_2 and T_2^*-weighted images reaching mesoscale resolutions within short acquisition times and with a high signal and contrast to noise ratio.

Main

Understanding the human brain, normal or diseased, is one the most significant scientific challenges of the 21st century, with academic, medical, societal and economic implications. In this context, neuroimaging, and especially Magnetic Resonance Imaging (MRI), has become an essential approach, as it allows maps of the brain anatomy, function, and structural connectivity to be obtained in patients and healthy volunteers *in situ* and non-invasively, as recognized by the Human Brain Project (now eBrain), the US Brain Initiatives and Brain Connects programs, and the China Brain Project [1]. Continuing efforts aim at boosting functional [2] and diffusion [3,4] MRI to explore the human brain *in vivo* at finer spatial resolutions [5]. By revealing at a mesoscopic scale the structural-functional links that compose the human brain (cytoarchitectonic of neuron clusters and their connections), one could bridge the gap toward both the extensive body of knowledge we already have at the microscopic scale on neurons and synapses, and the brain connectome at a macroscopic scale. This will allow us to confirm or refute our current hypotheses regarding its functioning and to generate new ones.

Benefitting from a supralinear gain in sensitivity [6,7], this goal can be reached by boosting the magnetic field, B_0, at which MRI systems operate. The Iseult project, originated in 2001, was based on this vision and aimed at building the first MRI magnet operating at 11.7T to investigate the human brain at an unprecedented scale. After nearly 20 years of research and development mainly led by the Department of Astrophysics, Particle Physics, Nuclear Physics and Associated Instrumentation (DAPNIA, since renamed IRFU) of CEA, the 132 tons 11.7T magnet (installation set-up shown in Figure 1) together with the scanner equipment delivered in 2021 its first images of the human brain *ex vivo* [8]. The drastic field homogeneity and temporal stability constraints for MRI were met respectively thanks to a careful design of the main superconducting coil (double pancakes) coupled with a passive iron shim, and an external power supply [9], reaching 0.9 ppm peak-to-peak over a 22 cm diameter sphere and 3 ppb/hour field drift at thermal equilibrium [8].

When increasing B_0, the radiofrequency (RF) field used to excite the nuclear spins becomes more and more inhomogeneous, a long-standing and major problem in MRI due to the shortening of the wavelength of the RF waves used (500 MHz at 11.7T for the hydrogen nucleus of water), leading to severe artifactual signal variations in the images visible already on the *ex-vivo* brain [8]. We tackled this problem by deploying parallel transmission hardware and software, first developed by the laboratory at 7T and further optimized for 11.7T (Methods). The specific absorption rate (SAR), also growing with B_o [10], was tamed with virtual observation points [11] and dedicated RF pulse design algorithms [12].

Approval from the regulatory body and ethics committee to perform first tests *in vivo* on 20 healthy adult subjects was granted early 2023. Overall, for small flip angle excitations and large flip angle refocusing pulses, normalized root mean square errors (NRMSE) equal or lower than 13% thereby could be obtained over the whole brain, the latter value being the intrinsic inhomogeneity achieved at 3T with volume coils [13]. Inversion pulses remained difficult to achieve given the architecture of the coil, leaving very localized artefacts despite reaching 8% of NRMSE. Static B_0 shimming was performed up to second order for each subject, using a brain mask and a quadratic programming approach yielding on average 82.7 Hz standard deviation over the brain (0.17 ppm). After B_0 and RF field mapping for each subject, following the RF pulse designs T_2 and T_2^*-weighted acquisitions were successfully performed by deploying the parallel transmission techniques (Fig. 2). The presented *in vivo* human brain images acquired at 11.7T thereby reveal good signal homogeneity consistent with the simulations, without severe RF field inhomogeneity artefacts. A 2D T_2^*-weighted acquisition was repeated with the same protocol (resolution, bandwidth, acquisition time, number of receive elements) at 3T and 7T on different volunteers. The comparison highlights the gain in signal:noise ratio (SNR) at 11.7T, which can be traded for higher spatial (or temporal) resolutions, enabling access to fine brain details. With the limited number of subjects authorized in this first protocol, and considering the problems induced by head motion, we could obtain good quality images with resolutions up to 0.19×0.19×1 mm^3 resolution (36 nl voxel volume) in about 5 min. Images indicate sufficient SNR to boost further spatial resolution or decrease acquisition time in the future.

Whether human beings could withstand 1.5-hour scans at 11.7T was also uncharted territory, while vestibular effects have been reported in mice [14], as well as mild genotoxic effects with repeated or chronic field exposures [15]. This first study, therefore, proceeded with care to gather reasonable evidence for the absence of harmful effects in more realistic conditions [16,17]. Physiologic (cardiac pulsation, arterial pressure), vestibular [18] and cognitive [19,20] measurements were implemented in the protocol to verify the innocuousness of the large magnetic field on humans. Those tests were performed on the 20 subjects scanned at 11.7T and on another cohort of 20 subjects with the magnetic field switched off, to disentangle any effect from a possible psychological bias (nocebo effect). The subjects of the 0T group were not informed of the absence of the magnetic field and received the same instructions as the volunteers scanned at 11.7T. Due to the absence of gradient coil vibrations at 0T, we mimicked the sound of MRI sequences with recordings played by a loudspeaker hidden at the back of the magnet. Genotoxicity was also assessed but only for the 11.7T group with blood samples drawn

before and after the 90 min exposure to perform intra-subject comparisons. Across all the tests carried out in this protocol, the statistical analysis revealed no significant differences related to field exposure (Methods).

The next steps will focus on development and implementation of motion correction tools [21,22] together with highly accelerated sequences [23,24]. The deployment of more efficient RF coils [25], more receive channels count and more powerful gradients [5] is also underway to further increase performance, especially to perform high resolution fMRI and tractography. The safety and imaging data gathered in this first study confirm the applicability of MRI at 11.7T: humans can safely tolerate such intense magnetic fields while image mesoscale resolutions can be achieved on the brain in reasonable time, provided the RF field inhomogeneity problem is mitigated to return optimal SNR. It brings to the fingertips of the neuroscience community an extraordinary opportunity to discover the brain at an unprecedented scale, to explore the biological mechanisms underlying our mental life and to better identify the mechanisms at work in neurological (e.g. Alzheimer) or psychiatric diseases (depression, schizophrenia).

Declarations

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Authors contribution

NB, FM, VG, AA, CLS, AM, CP, LS, MB, AV and DLB conceived and designed the study. NB, FM, VG, AA, CLS, AM, ML, CP, LS, MB and AV conducted the experiments. NB, FM, VG, AA, CLS, AM, ML, CP, LS, MB, AV and DLB performed the analysis and interpretation of the data. NB and DLB wrote the manuscript with input from all authors. DLB conceived and steered the Iseult project. NB is the current Iseult project pilot.

Competing interests

NB, FM, VG, AA hold several patents related to this work (pTx). AM is an employee of Siemens Healthcare. The other authors declare no competing interests.

Data availability

All data supporting the findings of this study are available within the paper and its Extended Method section.

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Methods

Data acquisition and processing

The MR sessions were 90 min long and targeted anatomical T_1, T_2 and T_2^* contrasts. A home-made parallel transmission RF coil (pTx) with dedicated virtual observation points for SAR monitoring was used [26]. The coefficient of variation (std/mean) of the B_1^+ field in static RF shimming mode was 45% over the whole brain and on average over the subjects. Increased power losses at 500 MHz combined with the pairing strategy of the 16 transmit elements to be fed by 8 RF power amplifiers made inversion pulses particularly difficult to design, despite reaching 8% NRMSE on average over the subjects (5% min, 12% max). This underlined the necessity to have high power RF amplifiers (here 2 kW per channel) and the importance of low cable losses, transmit efficiency, a high transmit channels count and to optimize channels pairing. The small flip angle excitation (NRMSE»8% over the 3D brain) and high flip angle refocusing (~13%) pulses were designed with the k_T-point [27] and GRAPE [28] approaches respectively, using an active-set algorithm [12] with simultaneous k-space optimization [29] and under explicit hardware (peak power, average power, maximum gradient slew rate) and SAR constraints. The GRAPE pulse was a universal RF pulse [28] and was designed offline on a database constituted of the 6 first subject field maps. The RF and DB_0 field maps were acquired for each subject for the pulse design and shimming using 2D interferometric turbo-FLASH [30] at 5 mm isotropic resolution and 3D multi-echo (TE = 1.6/3.5/6 ms) GRE acquisitions at 2.5 mm isotropic resolution respectively. Sequence parameters for the T_2-weighted variable flip angle turbo spin-echo acquisition were: resolution = 0.55 × 0.55 × 0.55 mm^3, TR = 6 s, TA = 13 min, GRAPPA = 3 × 2, TE = 301 ms, matrix = 400 × 400 × 320, bandwidth = 250 Hz/pixel. All acquisitions were performed with up to second order shimming using quadratic programming and computed brain masks. Comparisons obtained with similar data acquired at 3T and 7T revealed less than linear B_0 field dispersions with field strength, indicating more than satisfying results in terms of effective B_0 homogeneity at 11.7T in the brain.

Slice-selective “spoke” pTx pulses [31] were designed for the T_2^*-weighted 2D GRE acquisitions. The sequence parameters were FA = 27°, TR = 600 ms, GRAPPA = 2, TE = 20 ms, resolution = 0.2 × 0.2 × 1 mm^3, readout bandwidth = 40 Hz/pixel, TA = 4 min 20 s (2 spokes [32] for the axial acquisition) or TA = 8 min 30 s (TIAMO [33] for the sagittal acquisition). Another T_2^*-weighted 2D GRE acquisition with resolution = 0.19 × 0.19 × 1 mm^3 (2 spokes, TA = 5 min 16 s, same parameters as above otherwise) and a turbo spin-echo sequence of resolution = 0.3 × 0.3 × 1 mm^3 (TIAMO, TR = 7 s, TE = 48 ms, bandwidth = 130 Hz/pixel, echo train length = 9, GRAPPA = 4, TA = 4 min 26 s, 7 slices) were implemented on the last volunteer. Careful analysis of the sequences aided with field monitoring [34] and sequence adjustments were performed prior to the in vivo experiments to minimize field perturbations during the acquisitions [35]. For the 0.2 × 0.2 × 1 mm^3 high-resolution 2D GRE experiments, similar scans were performed at 3T and 7T on different subjects to visualize the gains brought by field strength (Fig. 2). Using a longer TE at 3T and 7T could increase the contrast to noise ratio but at the expense of SNR. The signal yet was robust

with respect to flip angle variations given the slow increase of T_1 versus field strength, making the Ernst angle relatively constant. Acquisitions were performed with 1Tx (body coil)-32Rx, 8Tx-32Rx and 8Tx-32Rx head coils at 3T, 7T and 11.7T respectively.

Biological protocol and results

This study (IDRCB 2022-A02321-42) was approved by the French “Agence Nationale de Sûreté et du Médicament” (ANSM) regulatory body and a national ethics committee. Written informed consent was obtained from all volunteers. In addition to the usual counter-indications for MRI which constitute exclusion criteria (pacemakers, implants, metallic objects, tattoos, claustrophobia etc), one notable inclusion criterion was the age of the volunteers, comprised between 18 and 40 years to reduce inter subject variability in genotoxicity test results. Twenty volunteers (24.2±4.8 year old, 8 males) were scanned at 11.7T and performed the tests mentioned below. Another set of twenty volunteers (23.6±6.0 year old, 7 males) was exposed to a 0T, nocebo, field and underwent the same tests (except genotoxicity). For the latter group, the environment was the same 11.7T scanner but with the magnetic field ramped down. All volunteers received the same instructions. To minimize the number of field ramps for the magnet, the study first included the 0T group, followed by the 11.7T one.

Anxiety of the participants was first evaluated with a questionnaire, whereby a score above 20 on the Hamilton scale constituted an exclusion criterion. The participants underwent cognitive (before, during and after the MRI exam) and vestibular (before and after) tests. The cognitive tests inside the MR scanner were performed with and without gradient noise to aim at isolating a potential impact of the loud acoustic noise on the results. Blood samples were drawn before and after the exams of the 11.7T group to conduct a genotoxicity analysis, subcontracted to an external and certified laboratory (GenEvolutioN, Porcheville, France), and see a potential effect due to exposure of the strong magnetic field [15]. Arterial pressure and cardiac pulsation were measured before, during and after the MRI exam.

Genotoxic tests: Genotoxicity testing is based on the detection of chromosome damage in human cells. An alternative to measuring structural aberrations in mitotic cells is to measure micronuclei. These are produced from whole chromosomes or acentric fragments that are unable to attach to the spindle at mitosis and appear during the next interphase as small darkly staining bodies adjacent to the main daughter nucleus. Cytochalasin B (Cyto-B), if added to cultures, inhibits cytokinesis (cell division) but not karyokinesis (nuclear division) resulting in the formation of binucleate cells [36] including micronuclei. Having shown that the repetitive exposure to 3T in MRI [15] induces mainly terminal deletions, the micronuclei protocol was performed with cytochalasin B addition at 68h after lymphocyte growth stimulation and with gamma irradiation (IRCM irradiation platform, CEA, Fontenay-aux-roses, France) as positive controls.

Behavioral tests: To assess participants' executive functions, we employed the Attentional Network Task (ANT) [19,20], a paradigm designed to evaluate attentional focus capacity. In a sequence of trials, participants were instructed to promptly indicate the direction (left or right) of a target arrow presented on a computer screen. The target arrow was flanked by additional arrows that could either all align in the

same direction as the target (congruent condition) or point towards the opposite direction (incongruent condition). Furthermore, in certain trials, advance cues provided information about the timing and/or location of the impending target.

Participants underwent this task on four occasions: prior to entering the scanner (Run "1_out"), within the scanner with no noise (Run "2_in"), inside the scanner with the added noise of an MR sequence (Run "3_in_noise"), and outside the scanner (Run "4_out"). Evaluations outside the scanner were conducted in a soundproofed room, with participants seated before a computer screen displaying the stimuli. Within the scanner, stimuli were visible through a mirror mounted on the head coil, with projections onto an LCD screen positioned at the rear of the scanner.

Balance tests: The participants' static balance was assessed before and approximately 25 minutes after the MR exposure (0T or 11.7T). A sensation questionnaire was filled after exiting the scanner in another room and before the vestibular test. The volunteers stood for 30s, first with their eyes open, then with eyes closed, on a force platform equipped with dedicated sensors and connected to a computer and software (AbilyCare, Paris, France) returning a stability score (ranging from 0 to 99) [18].

Results: In all tests performed throughout the protocol, no significant differences between the two groups exposed at 11.7T and 0T could be identified (extended data Fig. 3 and Table 1). No statistically significant differences in micronuclei count for genotoxicity for each subject before versus after exposure at 11.7T were found either. Follow up phone calls by the NeuroSpin medical team up to 1 week after exposure at 11.7T did not indicate any abnormality. Six/four out of twenty volunteers exposed to the 0T/11.7T field reported fatigue the same day or the day after. Four/one volunteers belonging to the 0T/11.7T group reported headaches. Notably different sensations experienced between the two groups involve transient dizziness when entering and exiting the magnet, as well as metallic taste in the mouth, which are known minor side effects occurring when moving in a magnetic field.

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Figures

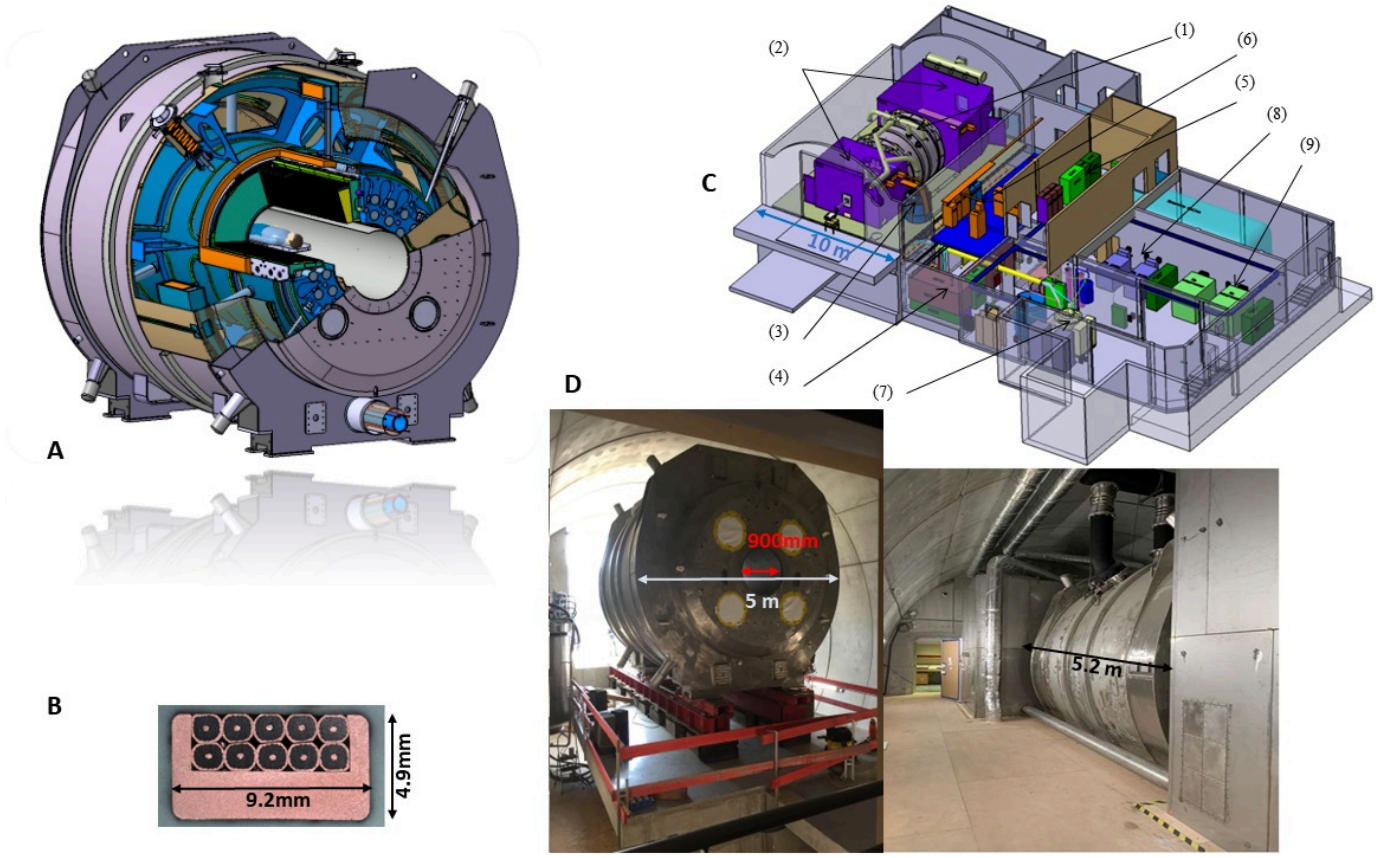

Figure 1

Iseult Magnet and installation set-up. *A: Schematic of the magnet showing the main coil and the 2 actively shielding coils. The colossal magnet weighs 132 tons, has an outer diameter and total length of just under 5 meters, the bore diameter is 90 cm ; B: cross-section of the wire used for the magnet showing the packing of the 10-strands of NbTi-Cu wire within a copper channel; C: General view of the Iseult MRI inside the NeuroSpin building (1): Magnet (2): Faraday cages (3): Satellite (4): Dump resistor (5):Magnet control command and Power Convertor (6): Siemens MRI equipment (7): Helium liquefier (8): Helium pumps (9): Compressors. To operate the 182 kilometers wire in superconductive, cryostable mode and provide a large safety buffer for MRI operation, the magnet is cooled at a temperature of 1.8 K (standard MRI magnets operating at 4 K) with a bath of 7000 l superfluid helium continuously maintained by the dedicated cryogenic plant. The magnet is connected to its power supply and the dedicated cryogenic installed underneath through a cryoline. D: Left: Initial installation of the magnet within a dedicated arch (10 m in diameter) at NeuroSpin; Right: current side view of the magnet within the arch, extending above the ceiling and under the floor of the patient side room, going all the way in the back of the arch.*

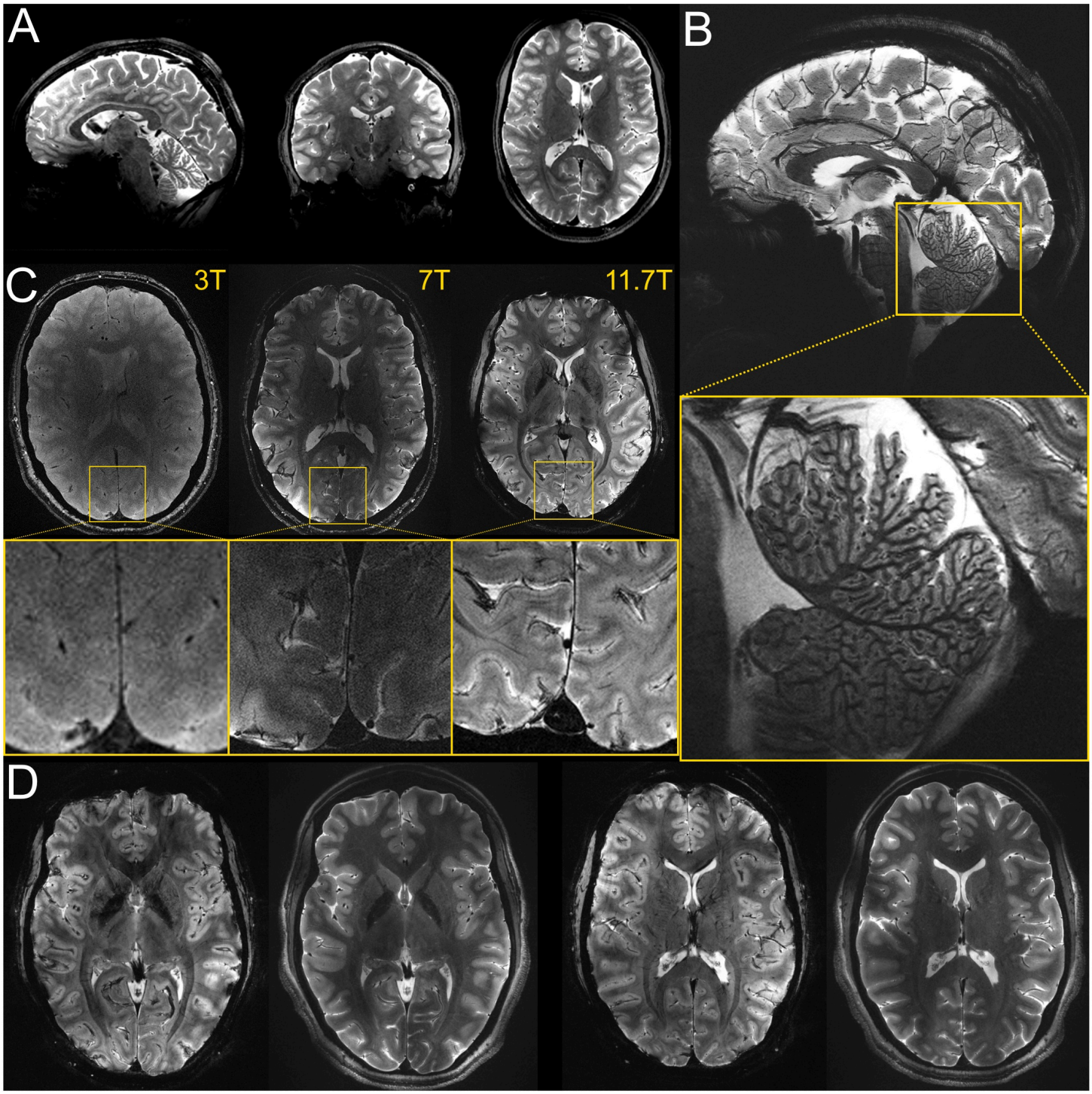

Figure 2

First in vivo images of the human brain obtained at 11.7T. *A: In vivo 3D T_2 variable flip angle turbo spin-echo acquisition at 11.7T with parallel transmission GRAPE universal pulses (resolution = 0.55 × 0.55 × 0.55 mm^3, acquisition time = 13 min) demonstrating the high B_1^+ (RF) field homogeneity reached throughout the whole brain volume. No severe RF field related artefacts are visible. B: In vivo T_2^*-weighted 2D GRE sagittal slice (resolution = 0.2 × 0.2 × 1 mm3, acquisition time = 8 min 30 s) showing*

striking details around the calcarine fissure and within the cerebellum. C: T_2^*-weighted 2D GRE axial images acquired at 3T, 7T and 11.7T (different subjects) (resolution = 0.2 × 0.2 × 1 mm^3, acquisition time = 4 min 20 s). Details within the cortical ribbon become clearly visible at 11.7T and not at lower field strength due to poorer signal and contrast to noise ratios. D: 11.7T T_2^*-weighted 2D GRE axial images (resolution = 0.19 × 0.19 × 1 mm^3, acquisition time = 5 min 16 s) juxtaposed to turbo spin-echo T_2-weighted images (resolution = 0.3 × 0.3 × 1 mm^3, acquisition time = 4 min 26 s).

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- NatMetANTResults.tif
- ExtendeddataTable1.docx